Design and Evaluation report

- ## Introduction

In this assignment, instead of making a web-based interactive experience using Artificial Intelligence or a simple web-based game, I chose and made audio-visual artwork. Audio-visual artwork like what did in semester 1, the difference between them is semester 2 artwork included sound. To get done the work, a range of research was undertaken, such as how to import audio to the sketch, more about p5.Amplitude and p5.FFT.

- ## Research

To complete the assignment, p5.js Sound Tutorial which is a playlist in YouTube created by The Coding Train is helpful, by watching all the videos, I learnt what is amplitude, how to import audio to the sketch and how to create a play button. Another word for volume is amplitude and if sound exists is some sort of waveform, the amplitude being the height of that wave the distance between the top and bottom of that wave defined by The Coding Train (2016). To increase the interactivity for the artwork, microphone input used as what learnt from the lecture, therefore the sketch in state 2 will change based on the sound from the microphone input. For state 1, the interactivity focused on the button created as The Coding Train (2016), when the button pressed, music will come out and the sketch pattern change by following the music.

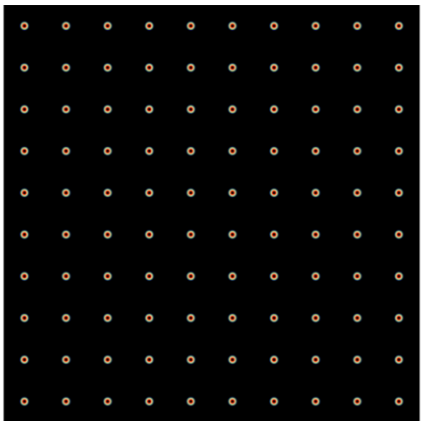
- ## Design and Inspiration

Most of the inspiration comes from lecture, state 2 based on Demo 3 from Creative Audio Visualizers (2024), state 1 based on The Coding Train (2016). From the inspiration, arrangement made on colour, number of loops, repeating shape and changing each colour to make sketch looks much more colourful.

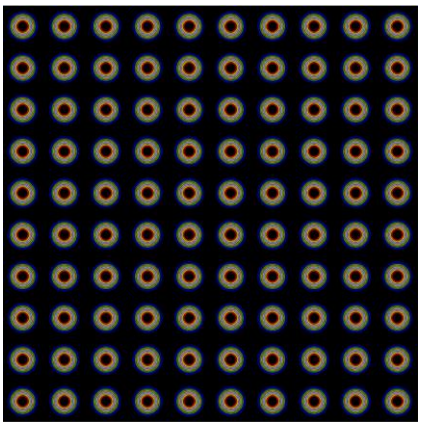
- ## Screenshots for Audio-visual artwork

State1

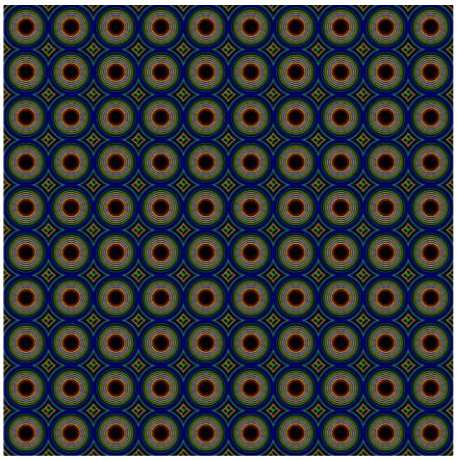
1.Before play the music



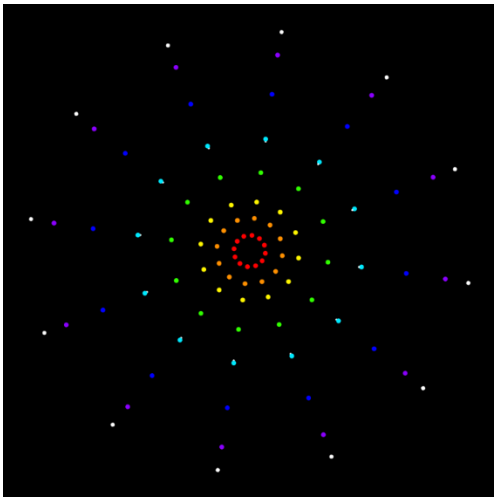
2. Beginning of music



3. middle of the music



State 2



- # Evaluation

The aim for the assignment is to create audio-visual artwork using p5.js. It involves analyzing audio input from the microphone and a loaded song to create dynamic visual patterns based on the audio frequencies. The visualization changes based on the mode selected, state 1 for mic input and state 2 for the loaded song.

Achievements from the assignment mainly have 2 parts, mic input and song visualization also song playback. Firstly, both parts also successfully capture input and visualize it as a dynamic pattern of circles and points, the visualization reflects changes in amplitude for state 1 and volume for state 2. State 2 visualizes mic input’s treble and bass frequencies as rotating points on the canvas while state 1 visualizes the song's amplitude as scale of ellipses. In addition, mapping of input data to visual parameters like scale and position is well-implemented. Secondly, implements song playback functionality with play and pause controls give interactivity to the work.

When doing the assignment, there are few challenges. Beginning with complex visualization logic, although reference other artist, but implementing the complex visual patterns based on audio frequencies required careful coding and testing. Next is mic input handling, efficiently handling mic input and mapping it to visual elements is the most challenging part. The last challenge is button functionality, it important for state 1 as it reflects the song's amplitude, therefore, is required to check if proper functionality of the play/pause button and its interaction with the song.

Most important for the assignment, for the code quality, code is modular with functions for mic input, song loading, and visualizations. Also, comments are provided throughout each line of code, aiding in understanding the logic and flow. In addition, variable names are descriptive, enhancing code readability. Lastly, loops for drawing visual elements are efficient and well-structured.

In conclusion, the work successfully achieves its goals of creating audio-visual artwork. It demonstrates effective use of p5.js libraries for audio analysis and canvas drawing. The visualization's responsiveness to mic input and song playback adds to its appeal. Going forward, improvements in handling mic input and possibly adding more visualization modes could enhance the project further.

- Link to my audio-visual artwork.

https://o2mok.panel.uwe.ac.uk/semester2/

- ## References

Milky Sky (2024). *Milky Sky. Soundsnap* [download] [Accessed 12 April 2024]

The Coding Train (2016). 17.2: Play and Pause Button - p5.js Sound Tutorial. *YouTube* [Video]. 08 June. Available from: https://www.youtube.com/watch?v=YcezEwOXun4&list=PLRqwX-V7Uu6aFcVjlDAkkGIixw70s7jpW&index=3 [Accessed 12 April 2024]

The Coding Train (2016). 17.4: Amplitude Analysis - p5.js Sound Tutorial. *YouTube* [Video]. 09 June. Available from: https://www.youtube.com/watch?v=NCCHQwNAN6Y&list=PLRqwX-V7Uu6aFcVjlDAkkGIixw70s7jpW&index=4 [Accessed 12 April 2024]

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Creative Audio Visualizers (2024). *Creative Audio Visualizers | Demo 3 | Codrops.* Available from: https://tympanus.net/Development/AudioVisualizers/index3.html [Accessed 01 April 2024]